

EDUCATION

Georgia Institute of Technology

Dissertation : Principled Sparsity for Efficient Deep Learning Across Computational Paradigms

Present

Ph.D. in ECE, Concentration in AI, Atlanta, GA

Aug 2021

- > Research in efficient AI, sparsity in LLMs, federated, multitask, and multimodal learning.
- > Supervised by [Dr. Vince Calhoun](#) and [Dr. Sergey Plis](#).
- > CGPA : 4.0/4.0
- > Expected Graduation : April 2026.

Georgia Institute of Technology

Thesis : Explicit Group Sparse Projection for Machine Learning

Dec 2021

Master's in ECE, Concentration in AI, Atlanta, GA

Aug 2019

- > Research in Sparse Neural Networks and Neural Network Pruning.
- > CGPA : 4.0/4.0

WORK EXPERIENCE

Google DeepMind

Mechanistic Interpretability and Model Alignment Team

Dec 2025

Research Intern, Kirkland, WA

Sep 2025

- > Conducting research on model and representation alignment, with a focus on controllability and interpretability.
- > Working with Diffusion Transformers and Image generation.

Cohere

LLM Efficiency Research

Dec 2024

Intern of the Technical Staff, Atlanta, GA

Sep 2024

- > Developed inference-time activation sparsity techniques for large language models (LLMs), enabling scalable and efficient inference when latency and cost are critical.

Dolby Laboratories

Experience Delivery Lab, Advanced Technologies Group (ATG)

Aug 2024

Ph.D. Research Intern, Atlanta, GA

May 2024

- > Designed and implemented an efficient fine-tuning method for LLMs using probabilistic layer selection.
- > Optimized GPU utilization by targeting LLM layers most relevant to downstream tasks.

Meta FAIR

Sparsity and Efficiency

Aug 2022

Research Scientist Intern, Menlo Park, CA

May 2022

- > Designed & implemented a git-like library for version control & model compression called weigit.
- > Weigit was integrated as part of the open-source [facebookresearch/fairscale](https://facebookresearch.github.io/fairscale/) library.
- > Research on extreme sparsity in deep learning models using signal processing based techniques (e.g. FFT and DCT) during training.

TReNDS Center at Georgia Tech

Ph.D. Research in Sparsity in Deep Learning

Present

Graduate Research Assistant, Atlanta, GA

Aug 2019

- > Working on sparse deep learning, efficient AI and its applications in federated, reinforcement, multi-task and multimodal learning.
- > Designed a new sparse projection algorithm : [TMLR](#), [ICLR-HAET](#).
- > Developed a novel sparse offline-RL method : [NeurIPS 2024 \(main-track\)](#), [NeurIPS-offlineRL](#).
- > Designed a novel communication efficient federated learning method : [arXiv](#).

</> TECHNICAL STRENGTHS

- > Deep Learning, Machine Learning, Computer Vision, Efficient AI.
- > Python, C++, Matlab.
- > PyTorch, JAX, NumPy, Pandas.
- > Linux, slurm, cluster computing, bash scripting.

RELEVANT COURSEWORK

- Statistical Machine Learning
- Linear Algebra
- Advanced Programming Techniques
- Information processing in Neural Systems
- Convex Optimization
- Advanced DSP
- Fourier Analysis
- Real Analysis

PROJECTS AND OPEN SOURCE CONTRIBUTIONS

WEIGIT : A GIT-LIKE NEURAL NETWORK MODEL-WEIGHT TRACKING LIBRARY

2022

github.com/facebookresearch/fairscale

- > Open source contribution, project was added as part of the open source fairscale library maintained by Meta AI FAIR.
- > Designed & implemented a git-like model weight tracking library for tracking the changes of model weights during training.

DRONE SIMULATION USING OPENGL AND OPENMPI

2019

github.com/riohib/UAV-Simulation-OpenGL-OpenMPI

- > A C++ implementation of flight simulation for a pack of drones following physics mechanics equations.
- > Graphics was rendered using OpenGL on C++.
- > Each drone physics was handled by a separate compute node and all drones were coordinated among nodes using OpenMPI.

ENF DATA ACQUISITION AND ANALYSIS :

2016

github.com/riohib/IEEE-SP-Cup-2016

- > Collected 10 hours of Electric Network Frequency (ENF) data from the national Power Grid.
- > Analyzed data using Fourier Analysis and classified with Support Vector Machines.

PROFESSIONAL ACTIVITIES

2023- Present **Conference and Journal Reviewer** : TMLR, TNNLS, NeurIPS

SELECTED PUBLICATIONS

- 2025 Samin Yeasar Arnob, Zhan Su, Minseon Kim, Oleksiy Ostapenko, **Riyasat Ohib**, Esra'a Saleh, Doina Precup, Lucas Caccia, Alessandro Sordoni. *Exploring Sparse Adapters for Scalable Merging of Parameter Efficient Experts*. **COLM 2025**.
- 2024 Samin Yeasar, **Riyasat Ohib**, Sergey Plis, Amy Zhang, Alessandro Sordoni, and Doina Precup. *Efficient Reinforcement Learning by Discovering Neural Pathways*. **NeurIPS, 2024**. [webpage](#).
- 2024 **Riyasat Ohib**, Bishal Thapaliya, Gintare Karolina Dziugaite, Jingyu Liu, Vince Calhoun and Sergey Plis. *Unmasking Efficiency: Learning Salient Sparse Models in Non-IID Federated Learning*. [[arXiv](#)]
- 2024 **Riyasat Ohib**, Bishal Thapaliya, Jingyu Liu, Vince Calhoun and Sergey Plis. *Efficient Federated Learning on distributed NeuroImaging Data*. **Frontiers in Neuroinformatics**. [webpage](#)
- 2023 **Riyasat Ohib**, Bishal Thapaliya, Jingyu Liu, Vince Calhoun and Sergey Plis. *Decentralized Sparse Federated Learning for Efficient Training on Distributed NeuroImaging Data*. **Neurips Medical Imaging Workshop, 2023**
- 2023 **Riyasat Ohib**, Bishal Thapaliya, Pratyush Reddy, Jingyu Liu, Vince Calhoun and Sergey Plis. *SalientGrads: Sparse Models for Communication Efficient and data aware Distributed Federated Training*. **ICLR Sparsity in Neural Networks workshop (SNN), 2023**. [webpage](#)
- 2022 **Riyasat Ohib**, Nicolas Gillis, Niccolo Dalmaso, Vamsi Potluru and Sergey Plis. *Explicit Group Sparse Projection with applications to Deep Learning and NMF*. **Transactions on Machine Learning Research (TMLR), 2022**. [webpage](#)
- 2021 Samin Yeasar, **Riyasat Ohib**, Sergey Plis and Doina Precup. *Single-Shot Pruning for Offline Reinforcement Learning*. **NeurIPS Offline Reinforcement Learning workshop, 2021**. [paper](#) [webpage](#)
- 2021 **Riyasat Ohib**, Nicolas Gillis, Sameena Shah, Vamsi Potluru, Sergey Plis. *Grouped Sparse Projection for Deep Learning*. **ICLR Hardware Aware Efficient Training workshop, 2021**. [paper](#) [webpage](#)
- 2018 **Riyasat Ohib**, Samin Arnob, Muhtady Muhaisin, Riazul Arefin, Taslim Reza and MR. Amin. *ENF Based Machine Learning Classification for origin of Media Signals: Novel Features from Fourier Transform Profile*. **Accepted at ICEECS 2018** presented on Nov 13-14, 2018.
- 2017 Samin Yeasar, **Riyasat Ohib**, and Muhtady Muhaisin. *Power line extraction process from Bangladesh grid and exploring ENF based classification accuracy using machine learning*. **IEEE R10HTC Conference, 2017**. [paper](#)
- 2016 **Riyasat Ohib**, Samin Yeasar Arnob, Md Sayem Ali, Rakibul Hasan Sagor, and Md Ruhul Amin. *Metal nano-particle enhanced light absorption in Ga-As thin-film solar cell*. **IEEE Asia-Pacific Conference on Applied Electromagnetics**, pages 89–93, 2016. [paper](#)

Efficient AI, Sparsity and Compression

May 2023

TReNDS Center, Atlanta, GA

Aug 2020

- > Developed a novel Group Sparse Projection algorithm for training sparse deep models. published in [TMLR](#), initial work at [ICLR HAET](#) workshop.
- > Developed a communication efficient method for Federated learning (FL) in the non-IID data regime.
 - > Preliminary work published at ICLR Sparse Neural Network Workshop and full work on [arXiv](#).
 - > Applications to Neuroimaging published in [Frontiers for Neuroinformatics](#).

Sparse Model Merging of Large Language Models

Mar 2025

Georgia Tech, collaboration with [MILA](#), Montreal, Canada, Atlanta, GA

Sep 2014

- > Introduced sparse adapters as simple yet highly effective parameter-efficient building blocks for modular architectures, outperforming LoRA and full fine-tuning in our setting.
- > Demonstrated superior post-merging performance by scaling sparse adapter merging across up to 20 NLP tasks, establishing state-of-the-art in-distribution results for modular adaptation.
- > Work published in [COLM, 2025](#).

Sparsity in Reinforcement Learning and efficient multi-task Learning in RL

Sep 2024

Georgia Tech, collaboration with [MILA](#), Montreal, Canada, Atlanta, GA

May 2021

- > Neural network pruning for offline and online Reinforcement Learning tasks before training. Preliminary work accepted at [NeurIPS workshop](#)
- > Full work accepted at [NeurIPS, 2024](#).

Predicting Location of Audio Recordings

Mar 2016

IEEE Signal Processing Cup : Team and Programming Lead,

Sep 2015

- > Predicted the location of recording of audio files, exploiting embedded background power signatures from nearby electrical power lines via machine learning techniques.
- > Led the undergrad Signal Processing Cup team to 11th rank worldwide and an Honorable Mention in IEEE Signal Processing Cup, 2016.